Cardiac spatial pipeline · integrated atlas and maturation

Scope:408,054 original spatial profiles, seven resolved cardiac lineages, four ages; unresolved/mixed profiles retained.

Geometry: antibody-guided segmentation contacts plus30/50/100µm centroid graphs screened against the tissue-territory mask. Contact is an operational shared boundary, not independently verified membrane contact.

QC: inherited broad identities, strict FB/EC subtype evidence, all-lineage nuclear/core marker audit, and a separately named conservative-mask sensitivity. Missing nuclear area alone never removes a CM.

Niches: stage-balanced graph-neighbor composition, no focal RNA or maturation inputs, review burden excluded from clustering features, seed stability and assignment uncertainty exposed.

Maturation: measured21-gene core/native indices plus four panel categories, equal-category relative score andMyl2-withheld score. No physiological maturity calibration is asserted.

Non-CMs: measured FB matrix, EC cycling/guidance, immune, mural, pericyte and epicardial proxy programs; inferred sprouting is excluded from primary mechanism models.

Inference: joint composition/program models, one common CM cohort per geometry, technical → anatomy → CM-state adjustment, paired contact/proximity comparisons,512 primary block-bootstrap draws.

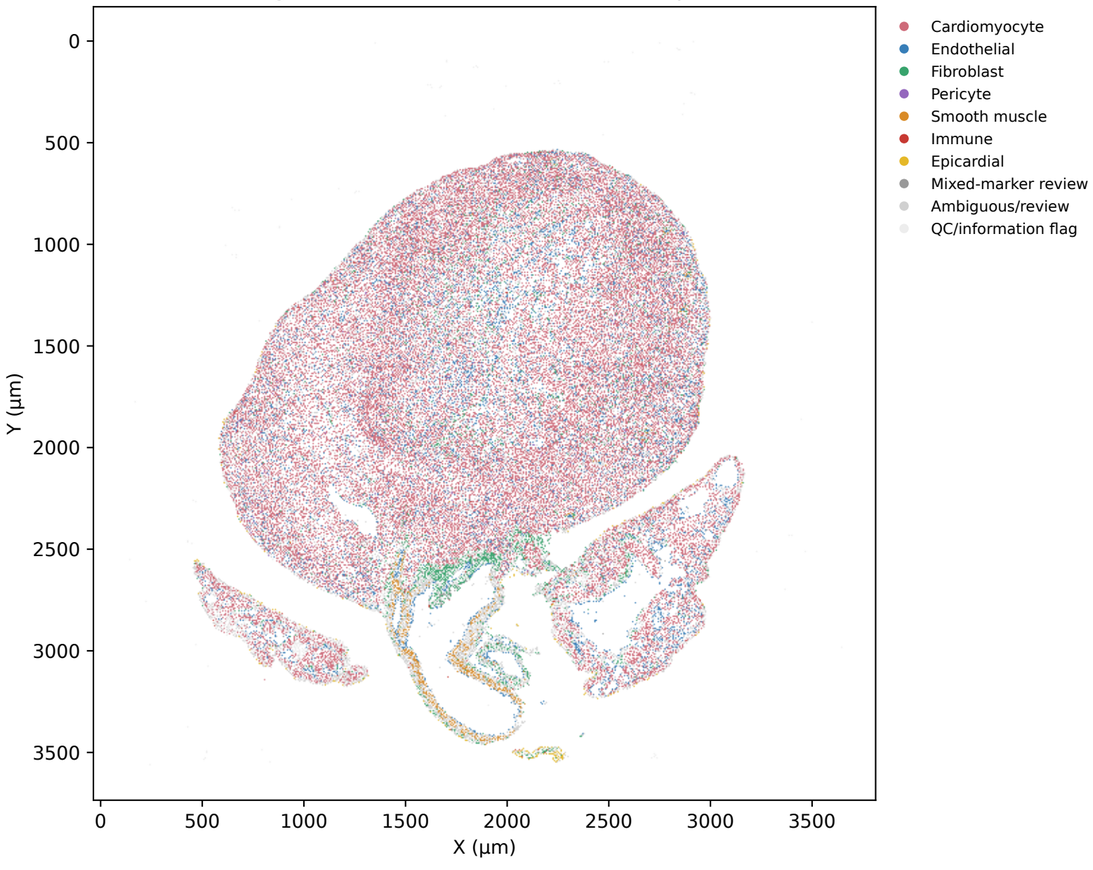
Prediction: spatial block holdout with150µm buffer for the primary50µm neighborhoods; compare baseline, composition, programs and joint predictors.

Method assessment: CellCharter is optional niche discovery; Wasserstein Wormhole is optional distribution embedding. This release uses neither package. It exports compatible inputs and runs a small exact/sliced OT distance pilot.

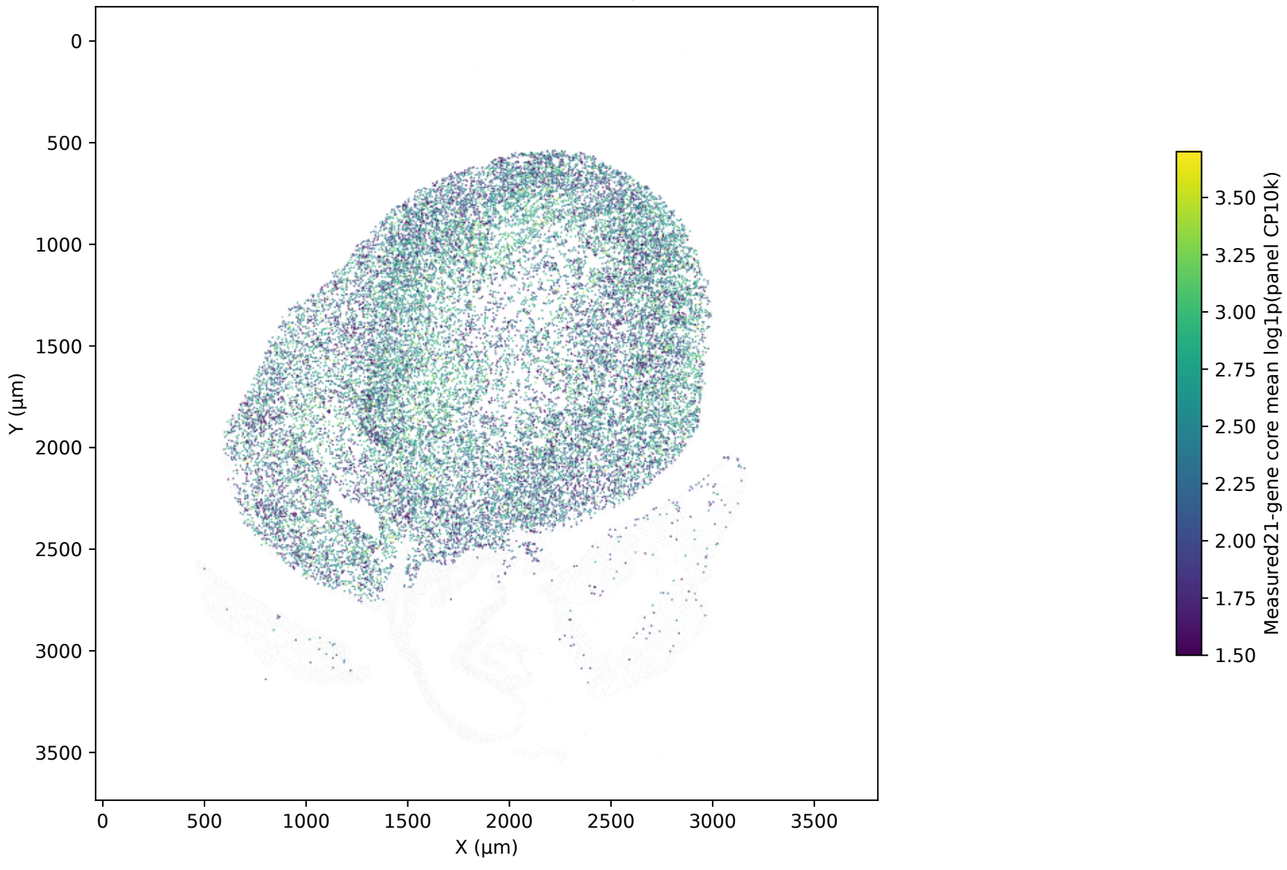
Biological scope: spatial association and candidate signaling hypotheses. One section per age cannot establish causal influence, longitudinal transitions or independently replicated age effects.

P0 · integrated spatial atlas, maturation, contexts and confidence

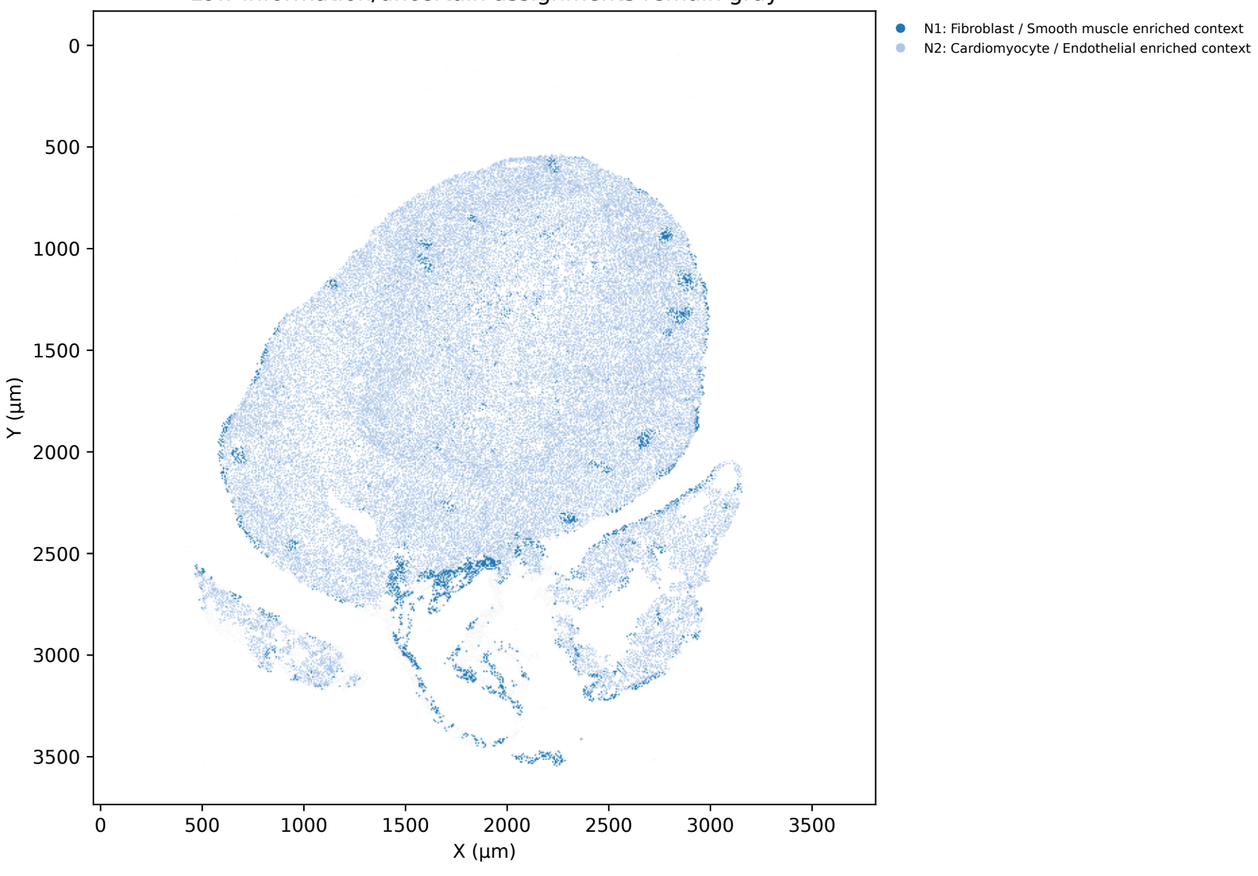
All seven lineages · inherited broad labels; review profiles retained



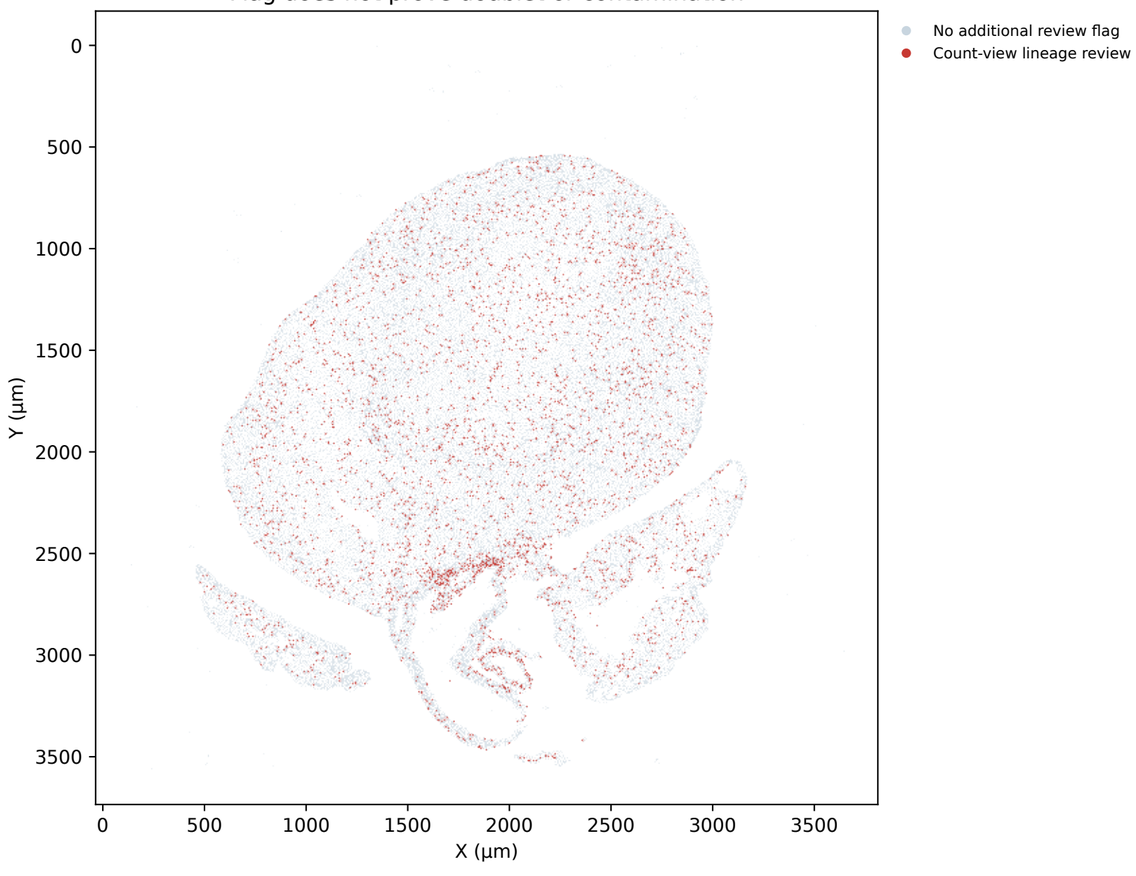
Ventricular CM maturation · boundary-core RNA



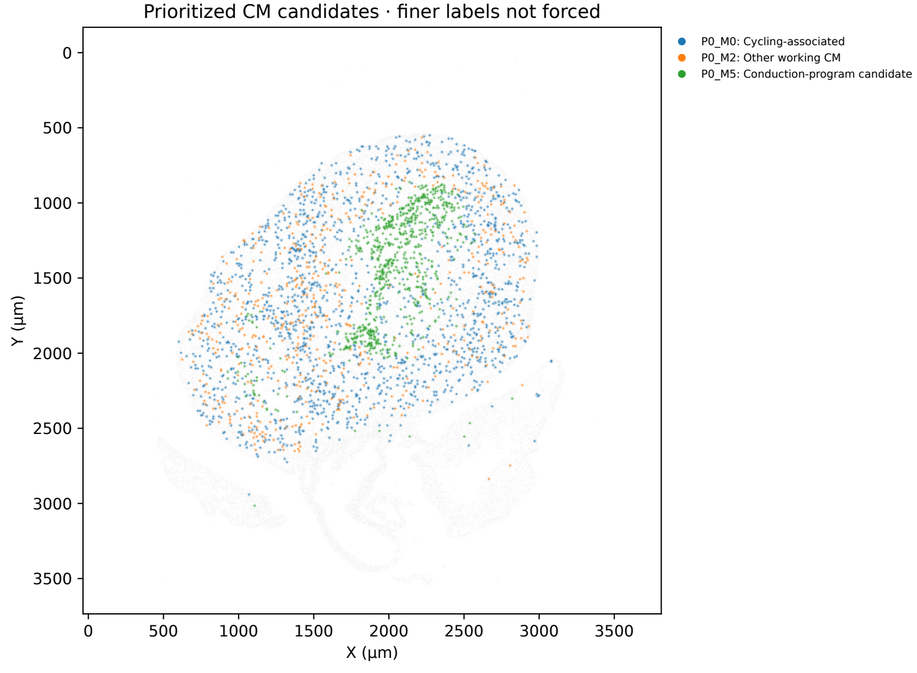
Outcome-blinded computational contexts
Low-information/uncertain assignments remain gray

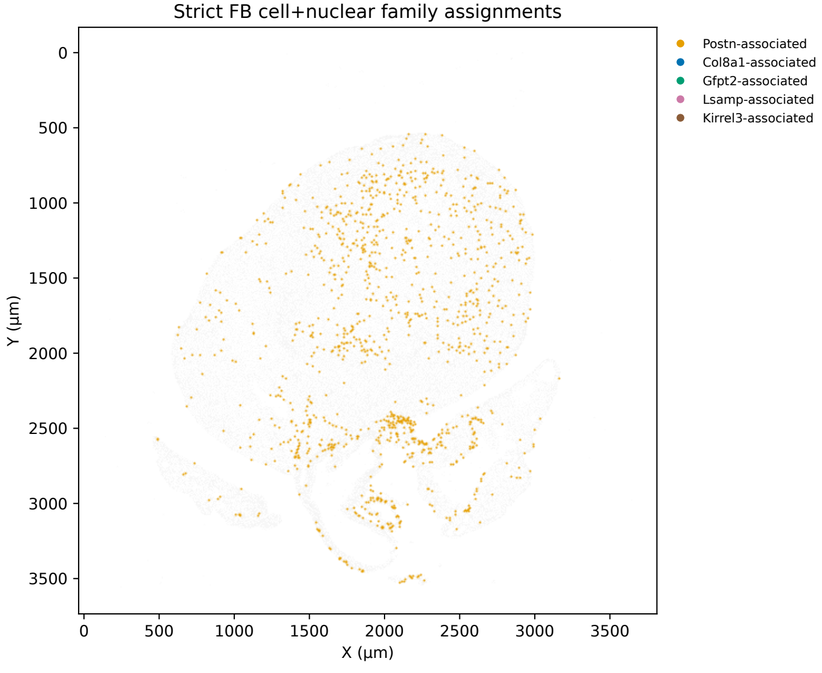


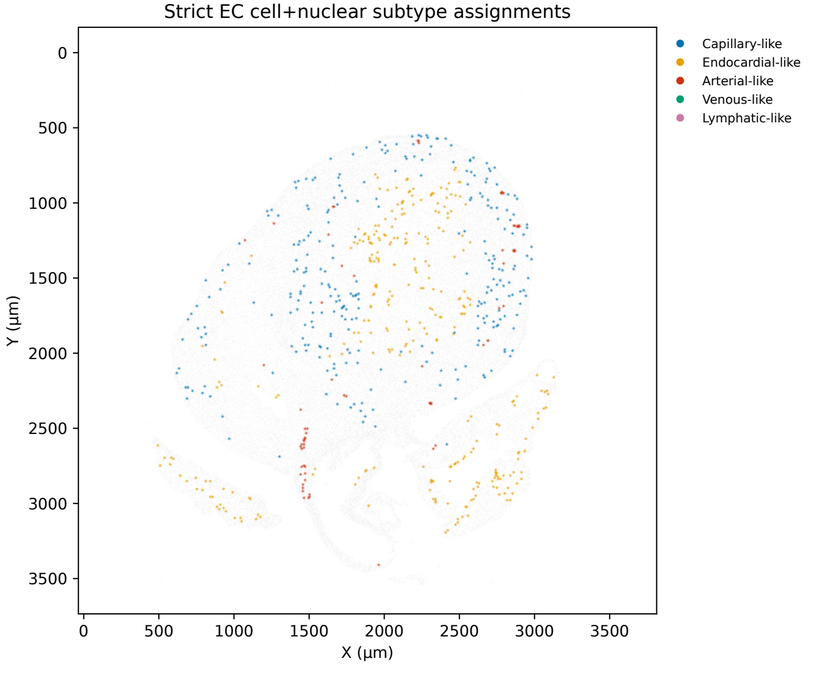
All-lineage QC · conservative review flags
Flag does not prove doublet or contamination



P0 · supported finer identities on original coordinates
Gray includes unresolved fine identity and other cardiac profiles; sparse categories are not absent-cell claims

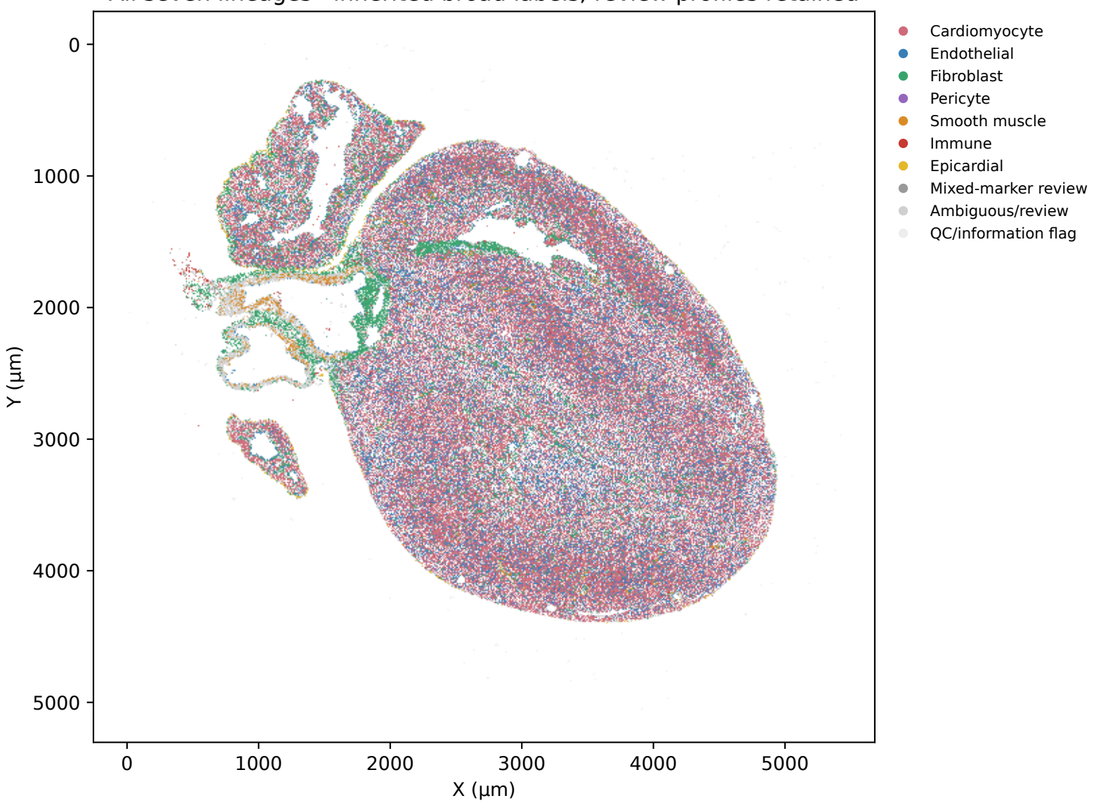




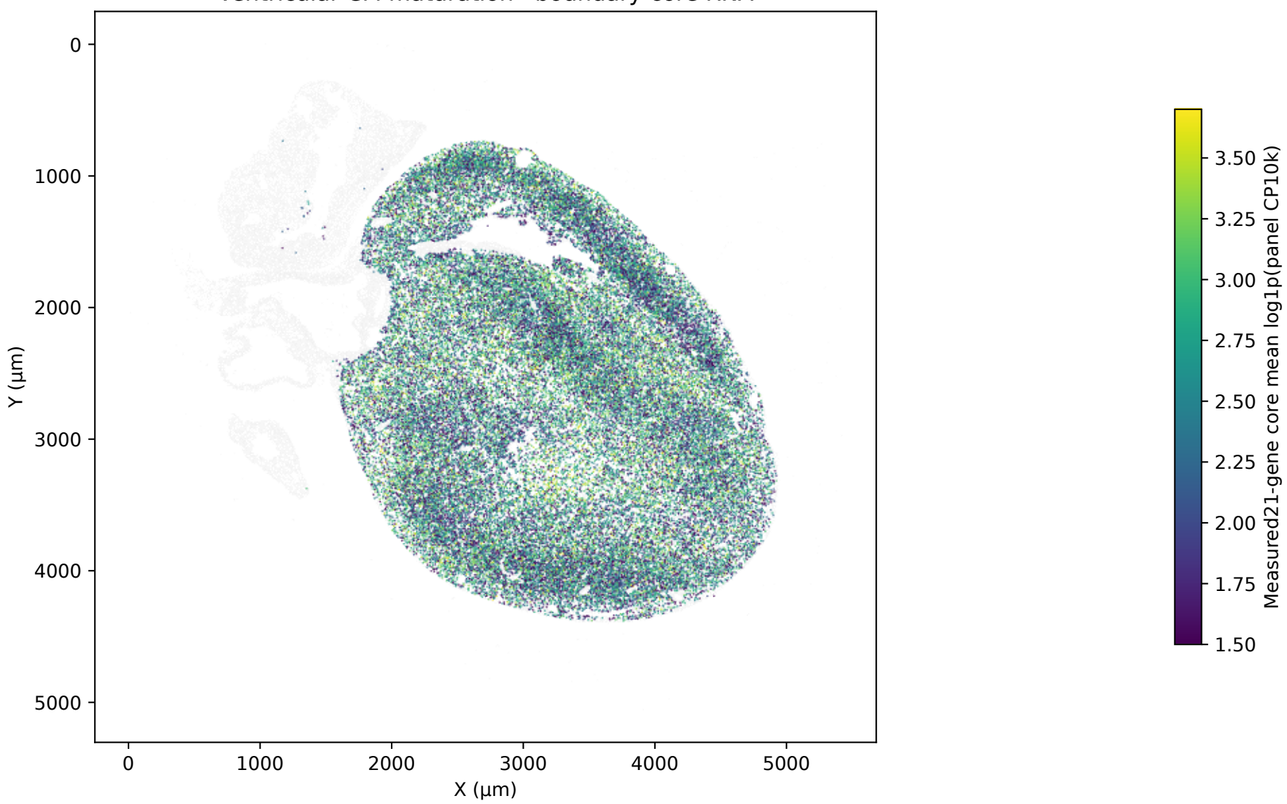


P7 · integrated spatial atlas, maturation, contexts and confidence

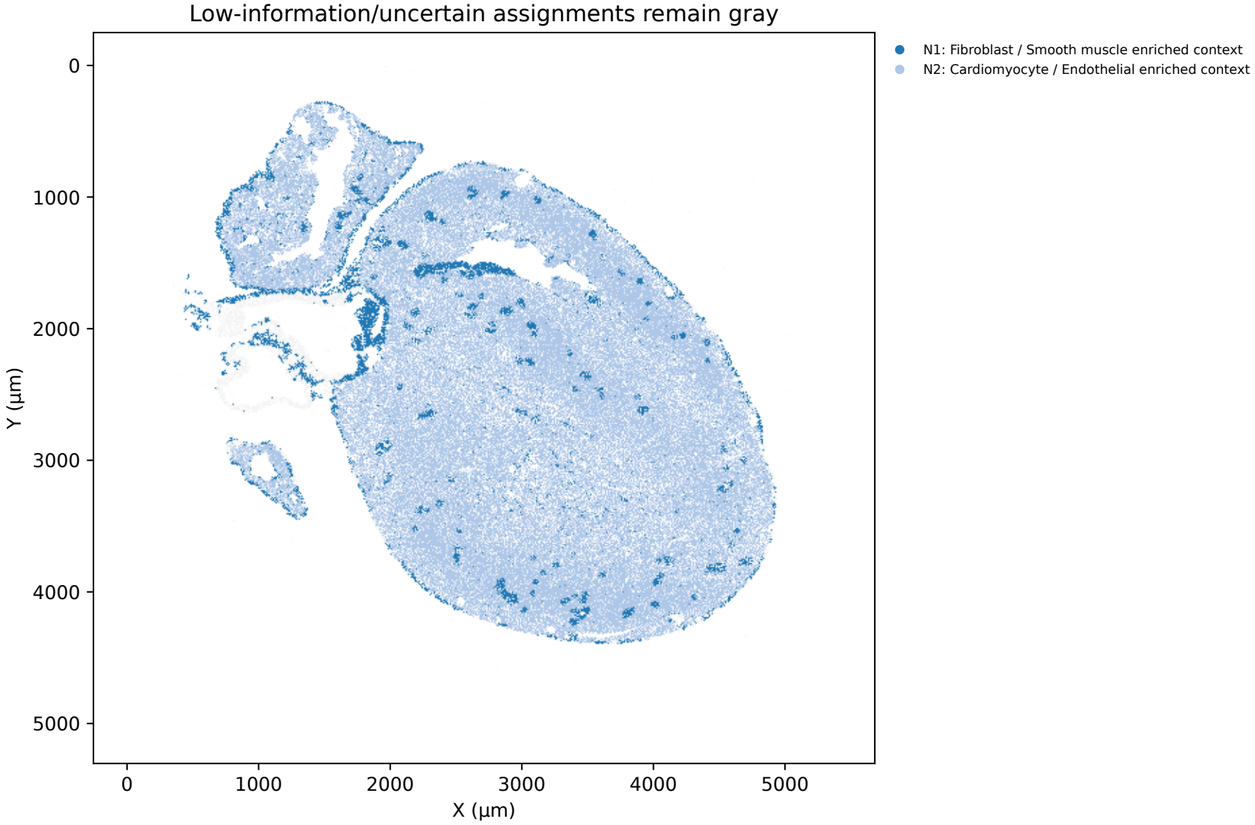
All seven lineages · inherited broad labels; review profiles retained



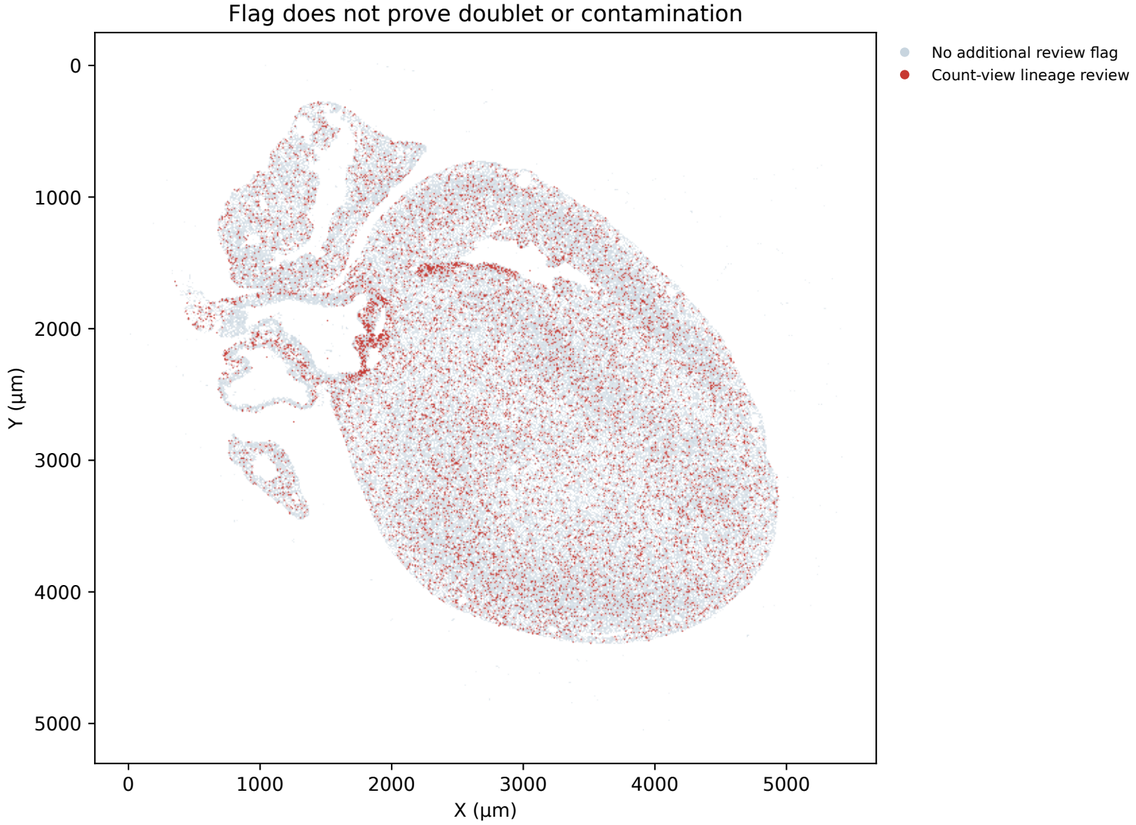
Ventricular CM maturation · boundary-core RNA

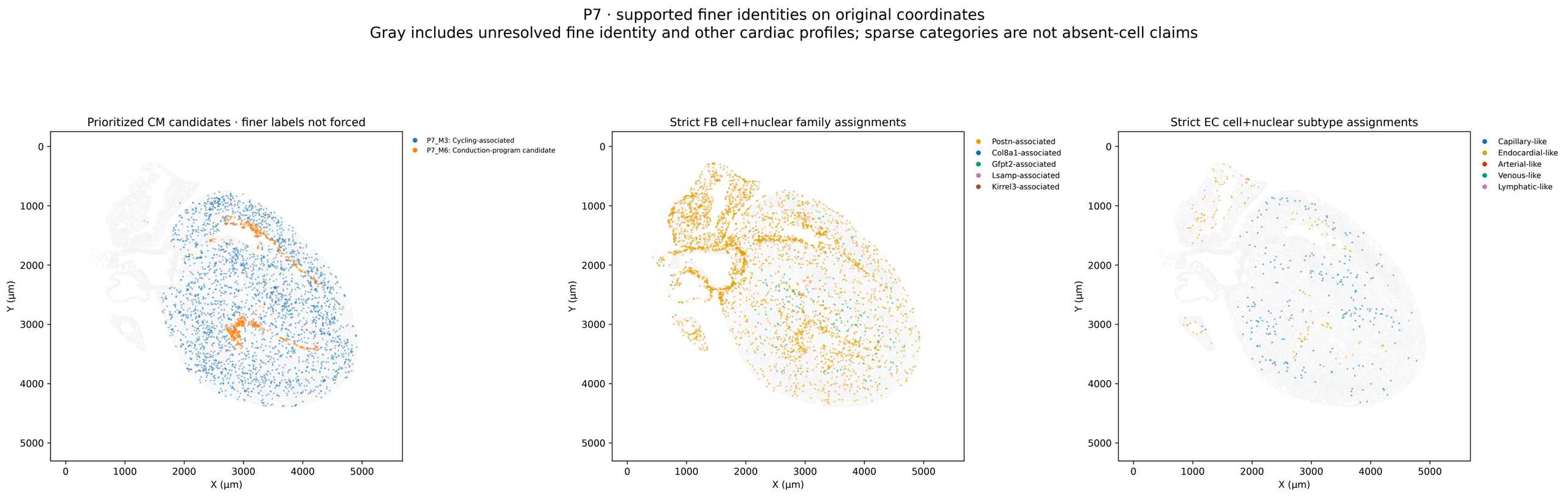


Outcome-blinded computational contexts
Low-information/uncertain assignments remain gray



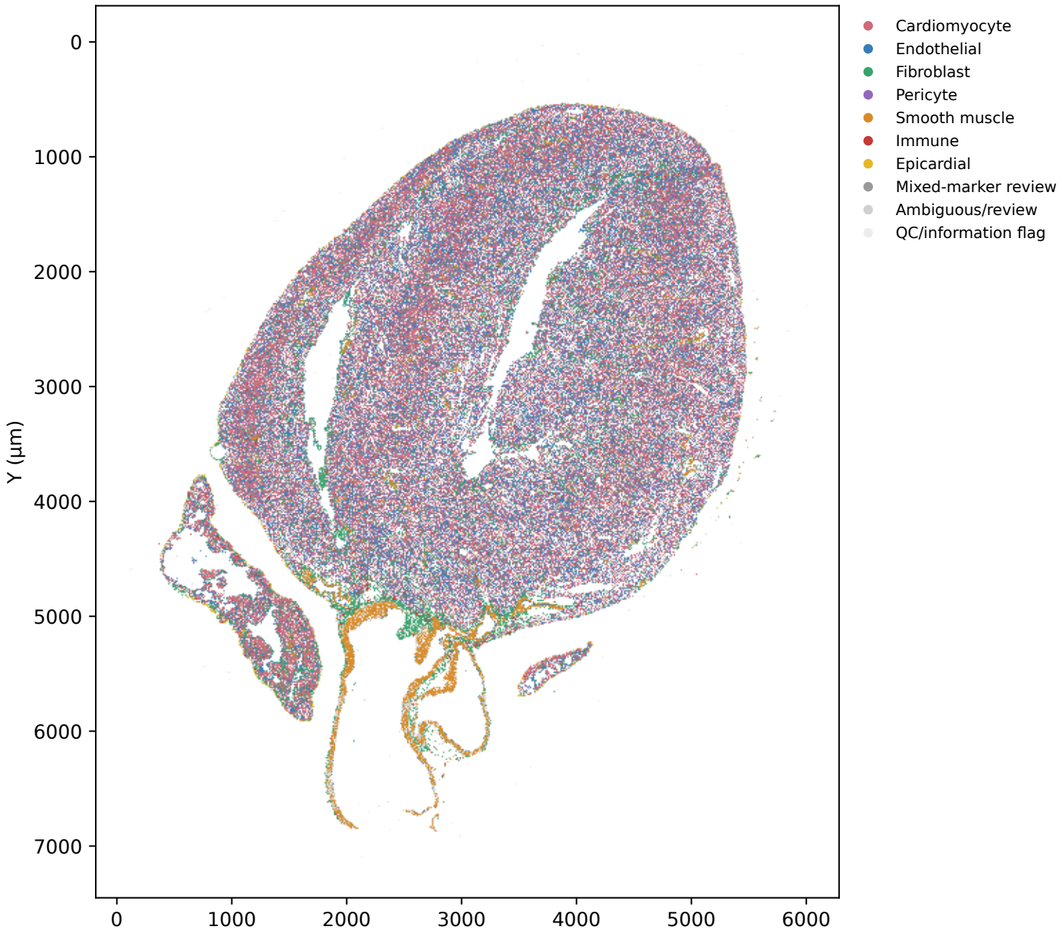
All-lineage QC · conservative review flags
Flag does not prove doublet or contamination



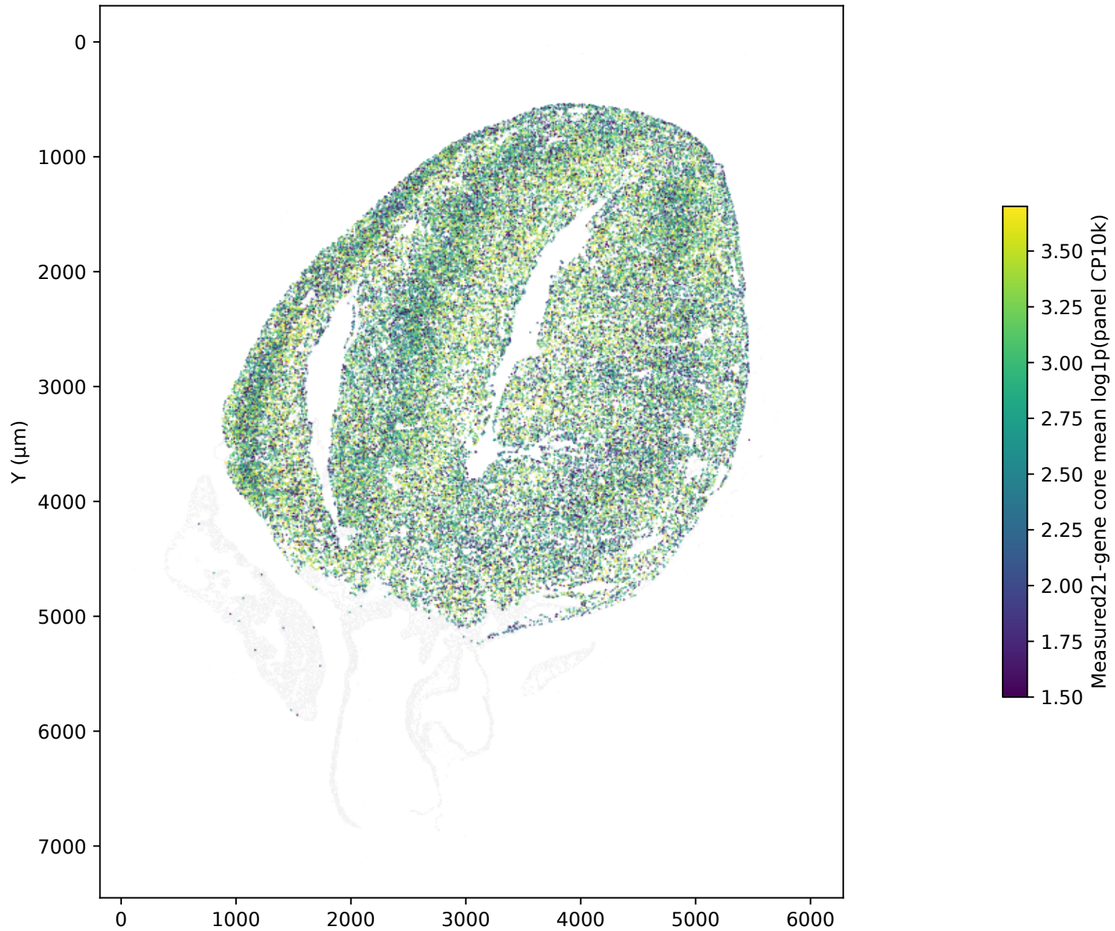

P7 · supported finer identities on original coordinates
Gray includes unresolved fine identity and other cardiac profiles; sparse categories are not absent-cell claims
Prioritized CM candidates · finer labels not forced
P7_M3: Cycling-associated
P7_M6: Conduction-program candidate
Strict FB cell+nuclear family assignments
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Strict EC cell+nuclear subtype assignments
Capillary-like
Endocardial-like
Arterial-like
Venous-like
Lymphatic-like
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000

P14 · integrated spatial atlas, maturation, contexts and confidence

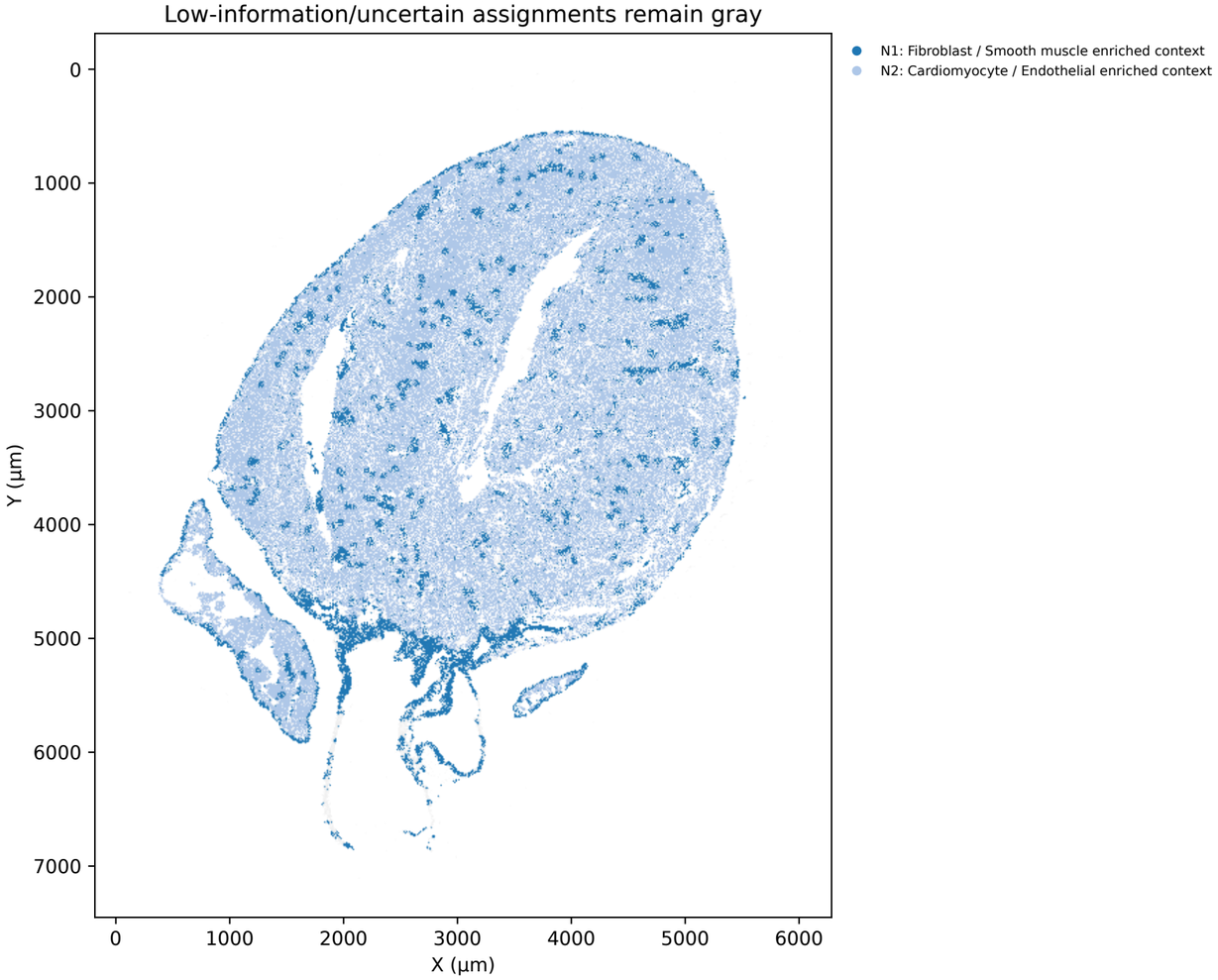
All seven lineages · inherited broad labels; review profiles retained



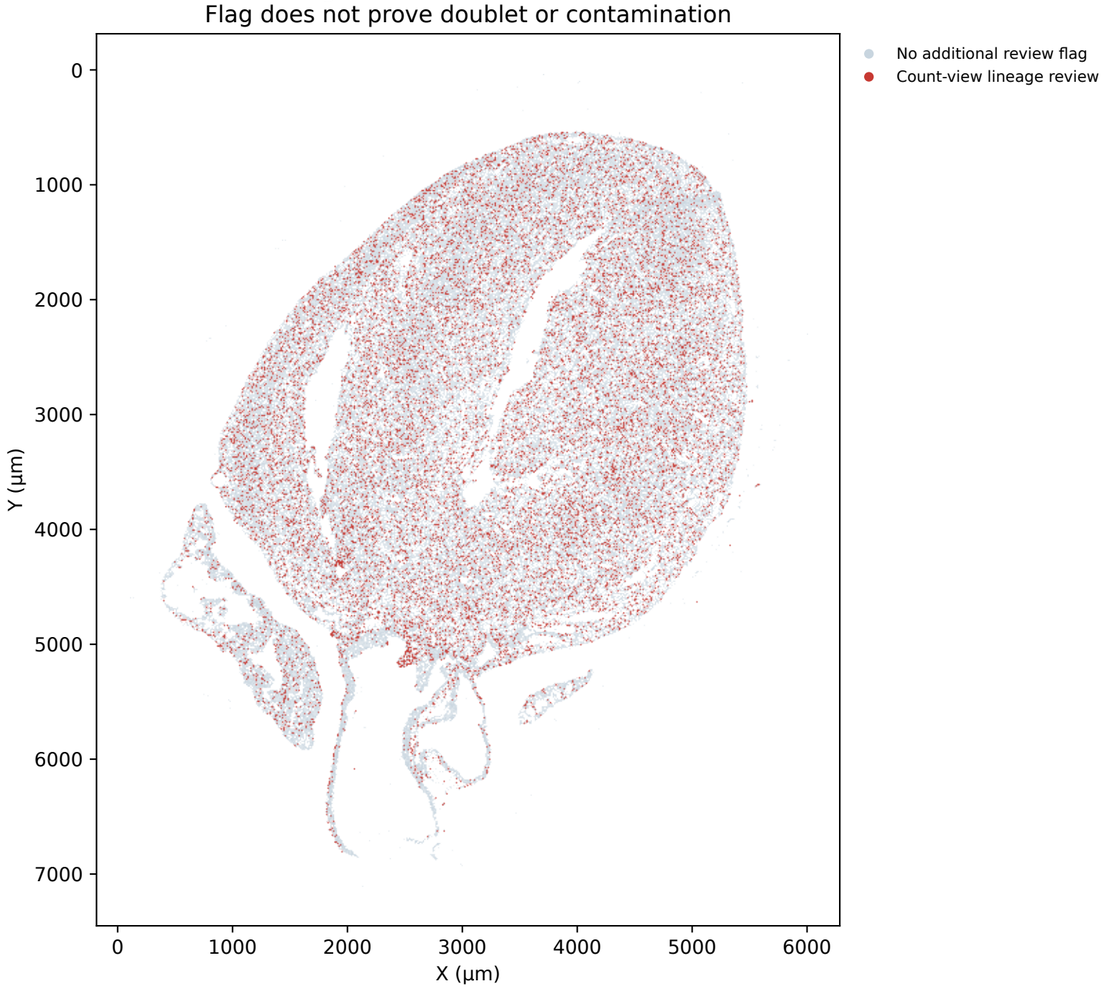
Ventricular CM maturation · boundary-core RNA

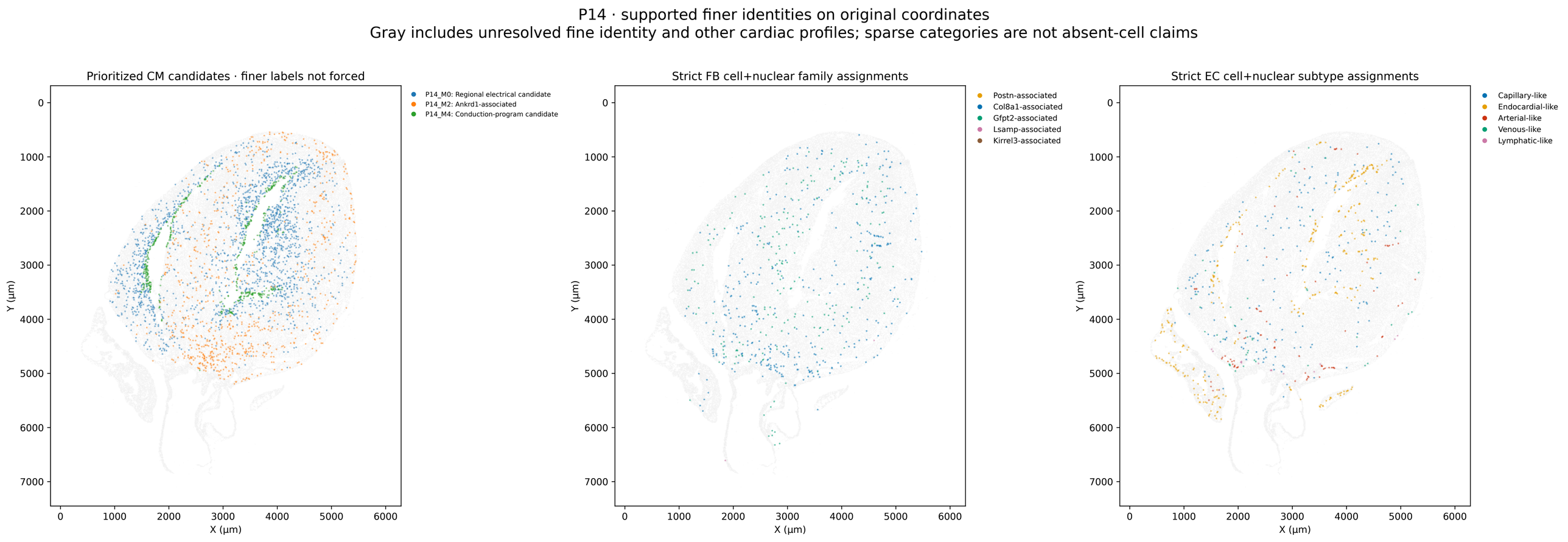


Outcome-blinded computational contexts
Low-information/uncertain assignments remain gray



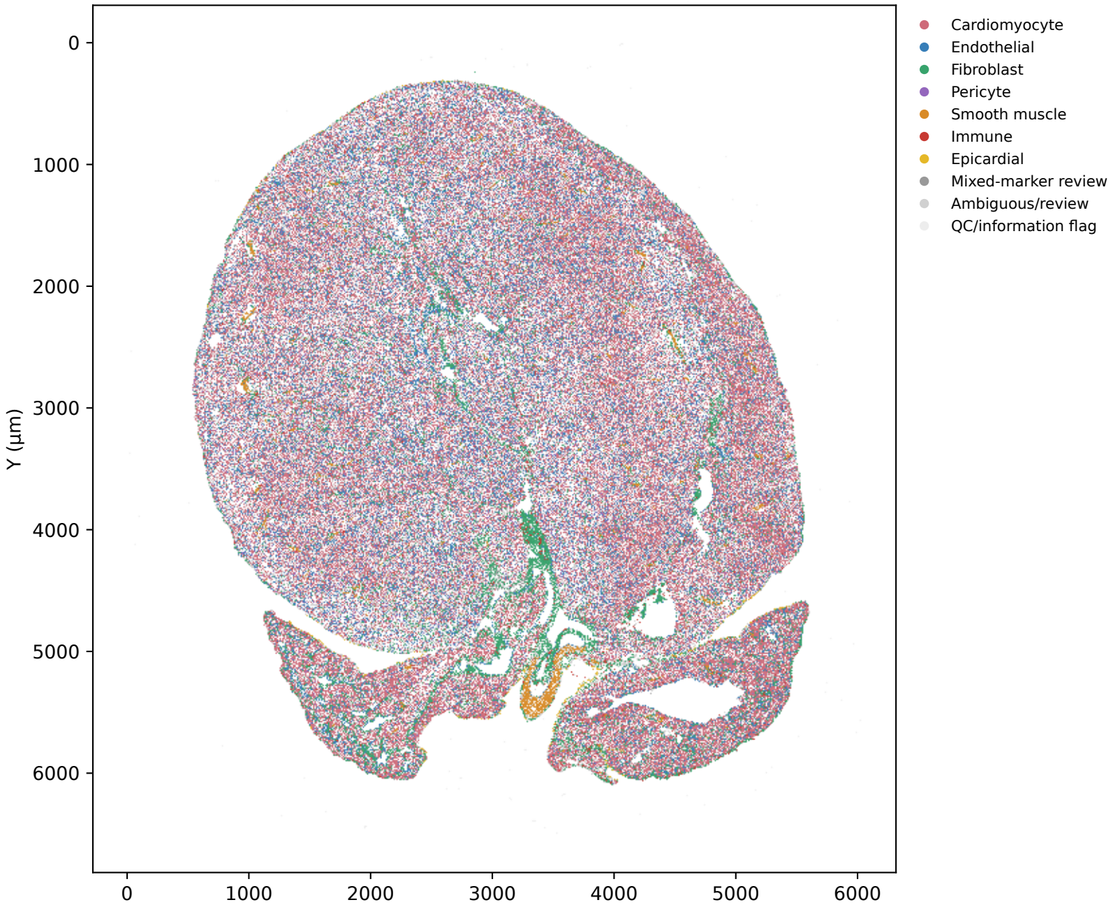
All-lineage QC · conservative review flags
Flag does not prove doublet or contamination



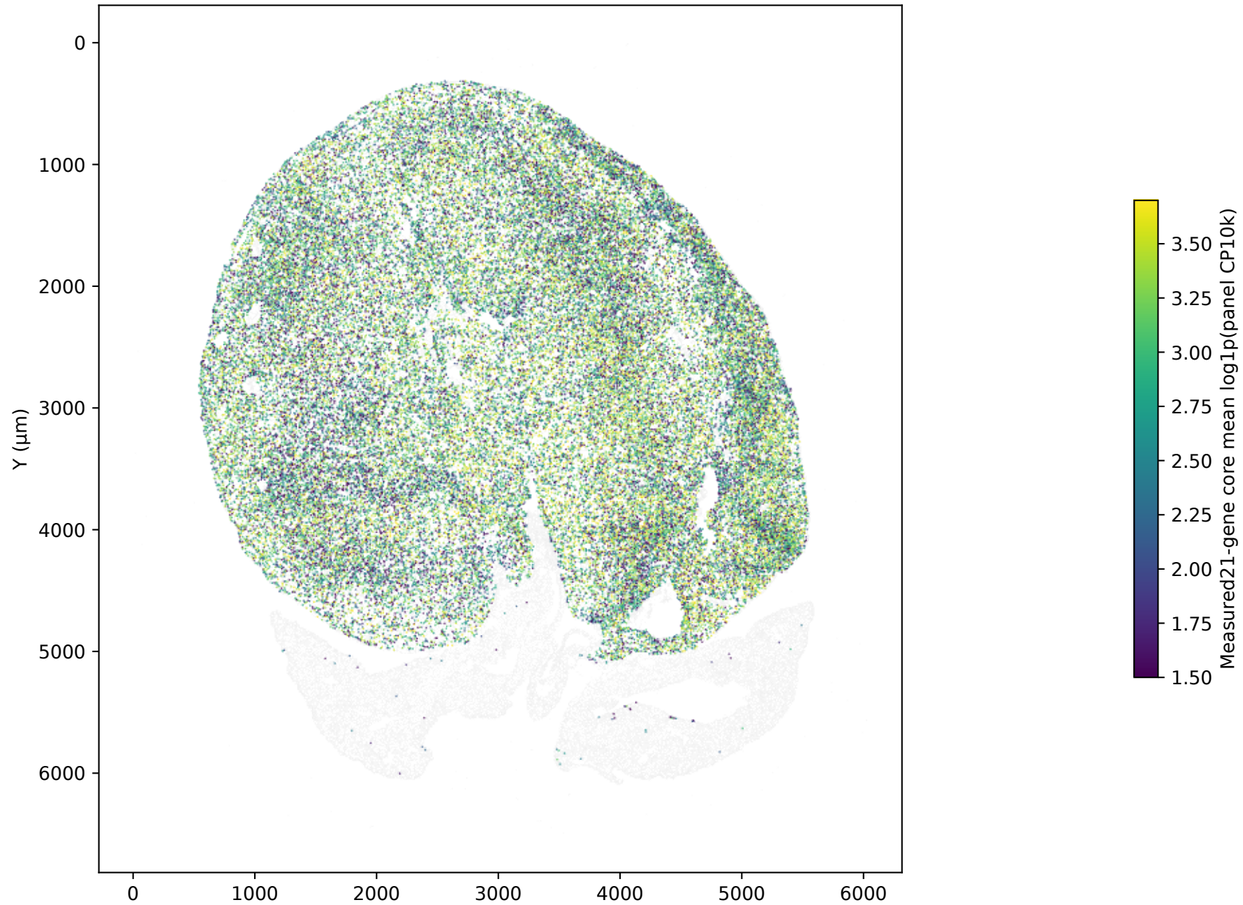

P14 · supported finer identities on original coordinates
Gray includes unresolved fine identity and other cardiac profiles; sparse categories are not absent-cell claims
Prioritized CM candidates · finer labels not forced
P14_M0: Regional electrical candidate
P14_M2: Ankrd1-associated
P14_M4: Conduction-program candidate
Strict FB cell+nuclear family assignments
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Strict EC cell+nuclear subtype assignments
Capillary-like
Endocardial-like
Arterial-like
Venous-like
Lymphatic-like
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000
6000
7000

P21 · integrated spatial atlas, maturation, contexts and confidence

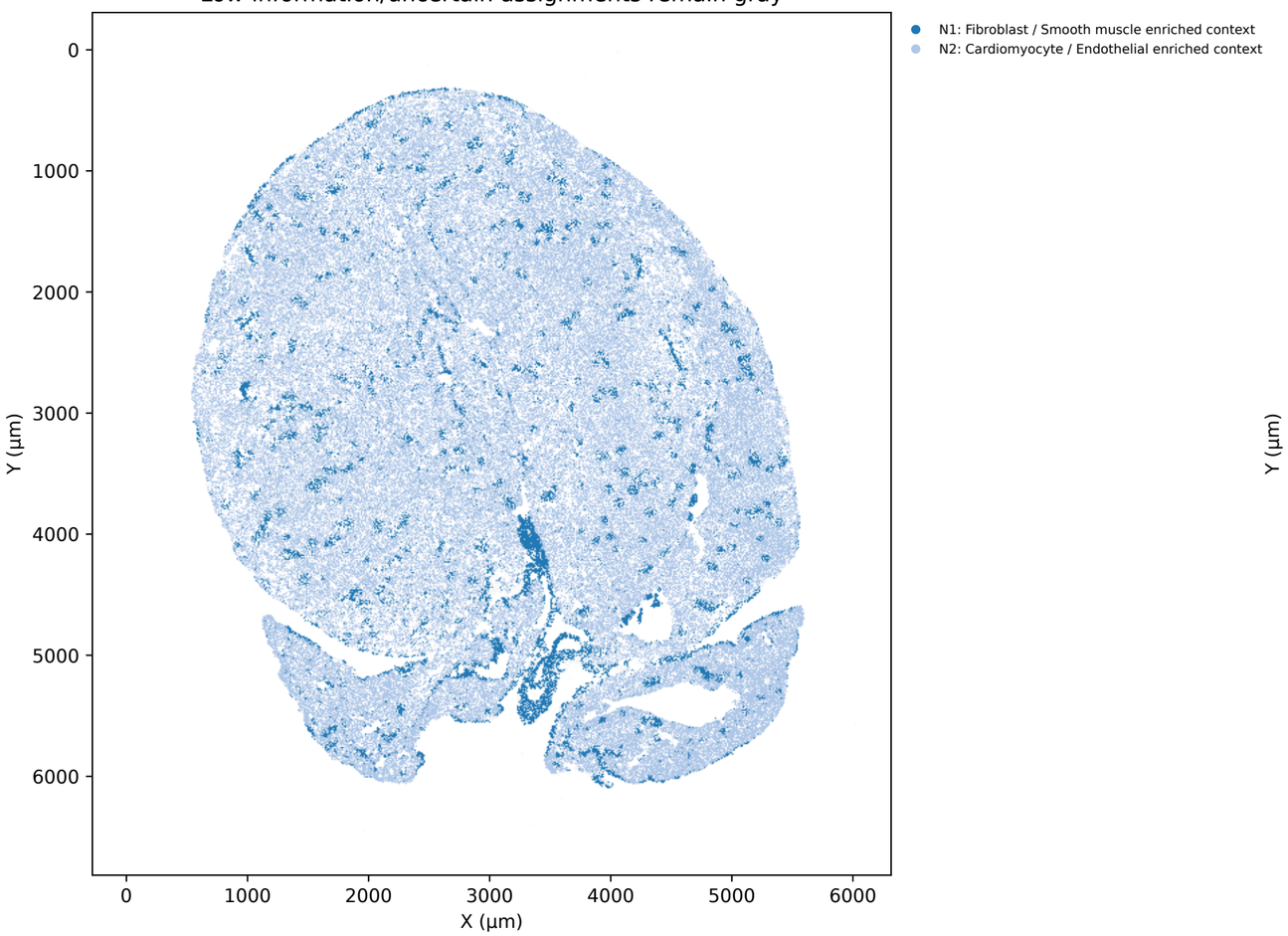
All seven lineages · inherited broad labels; review profiles retained



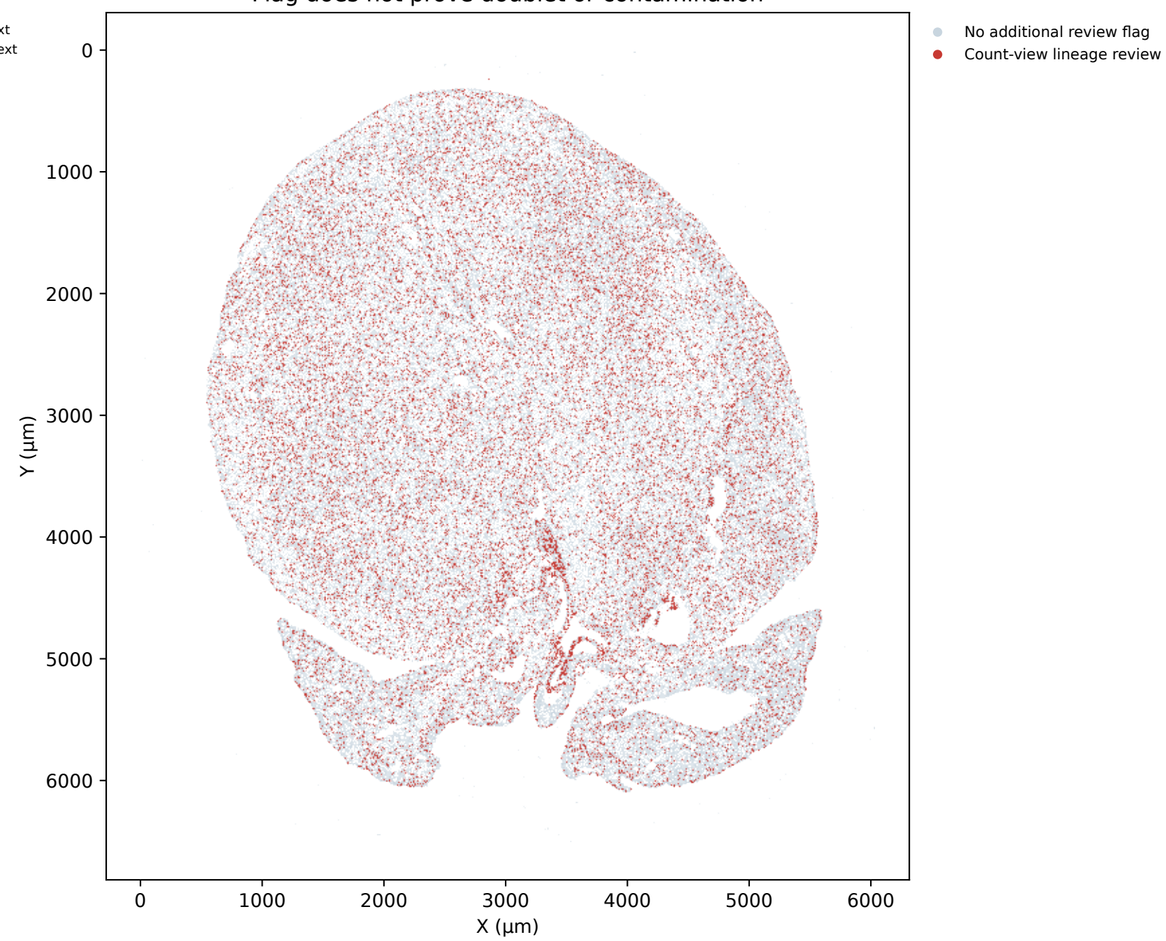
Ventricular CM maturation · boundary-core RNA

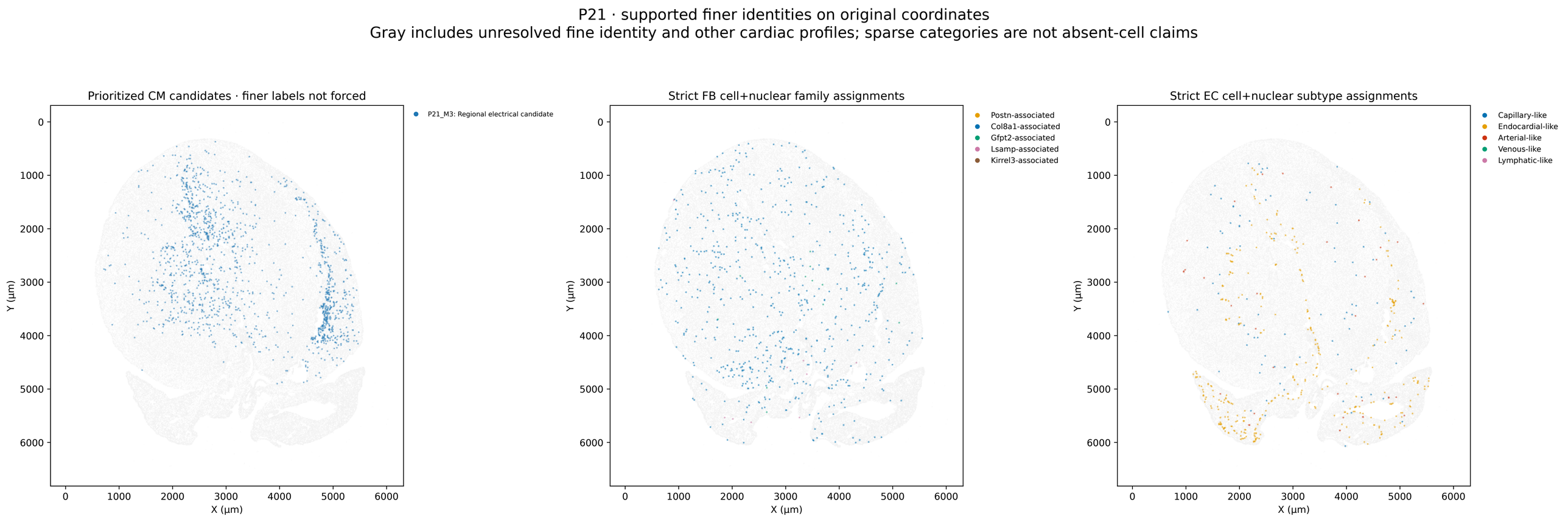


Outcome-blinded computational contexts
Low-information/uncertain assignments remain gray

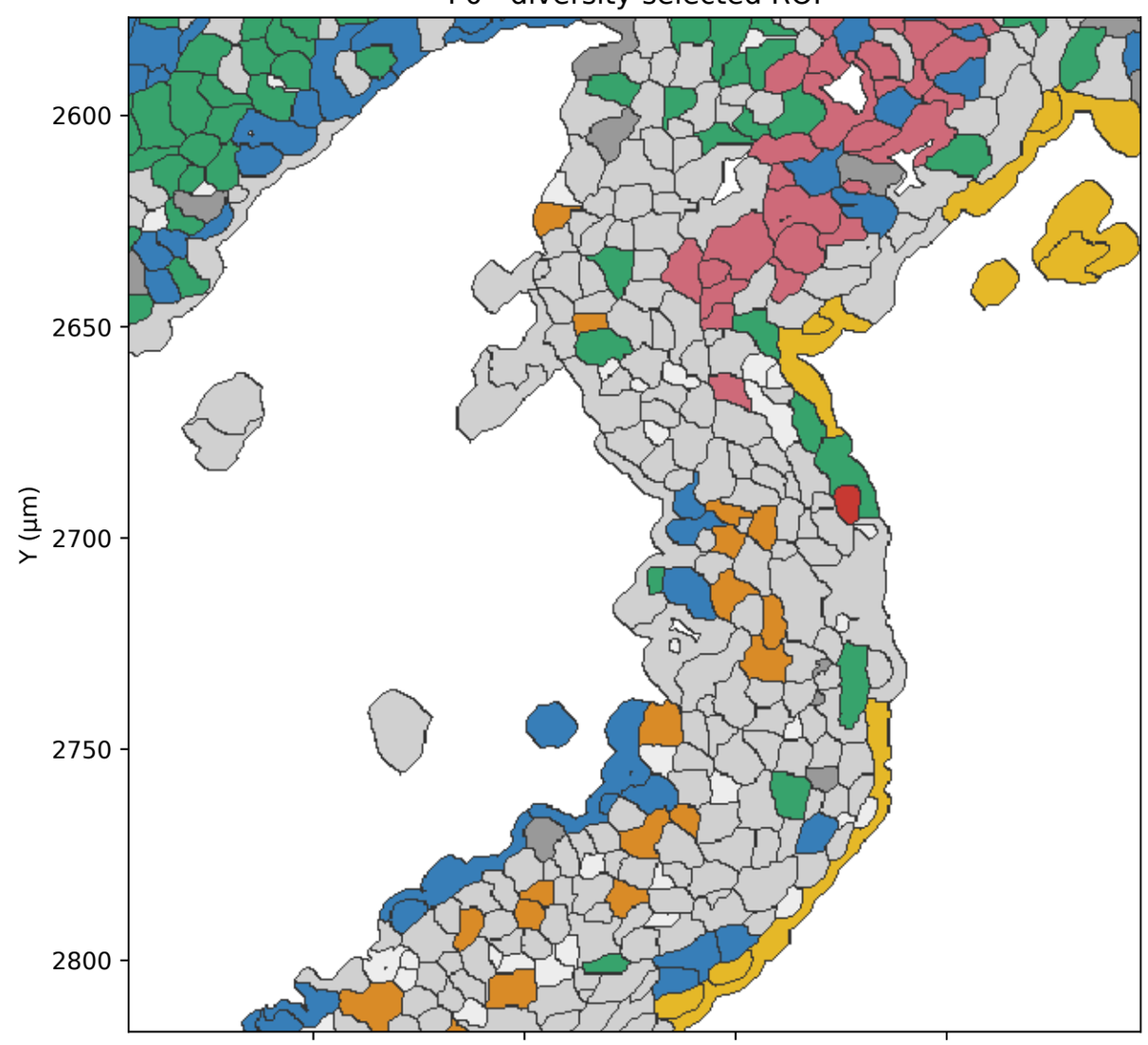


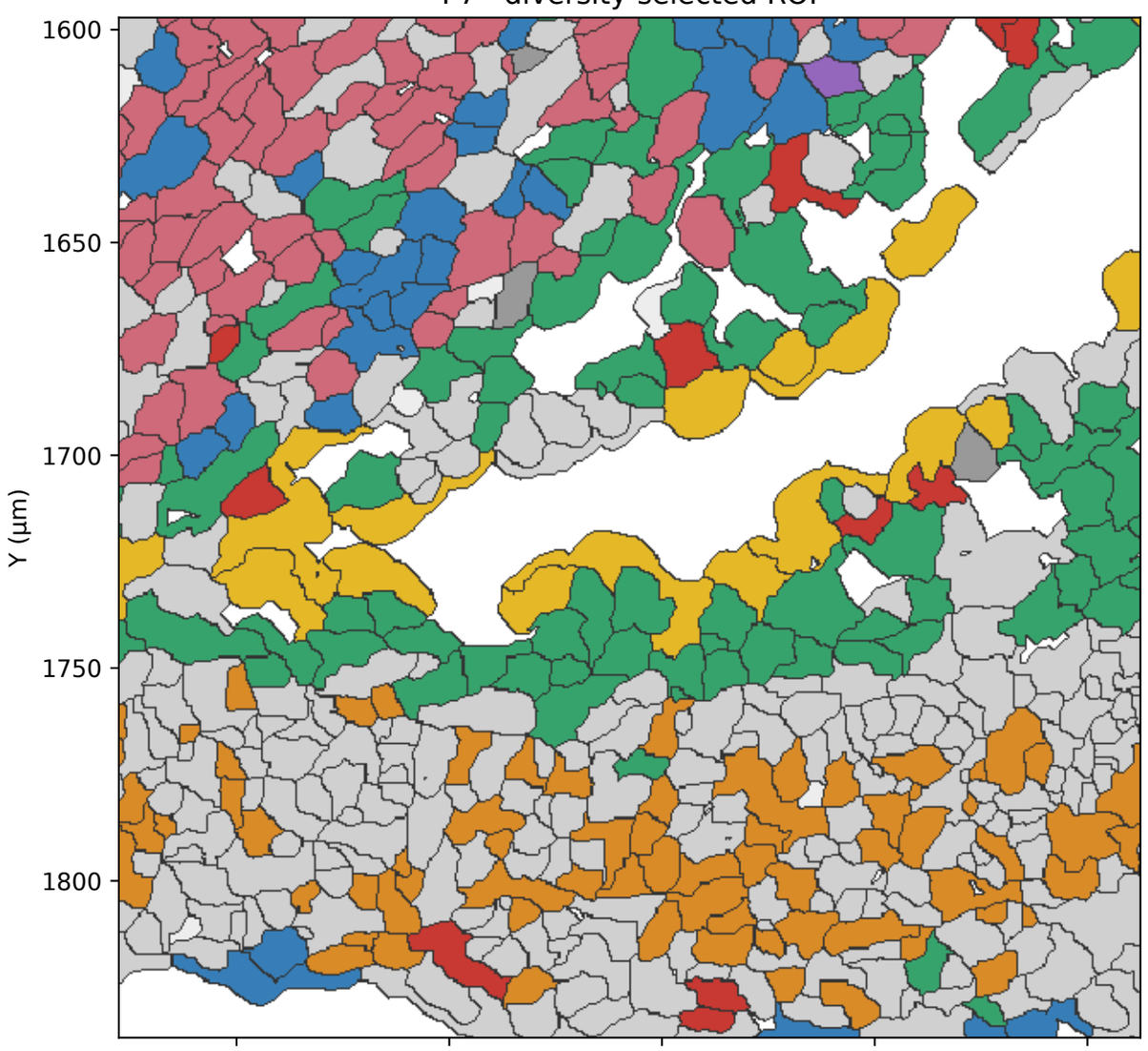
All-lineage QC · conservative review flags
Flag does not prove doublet or contamination

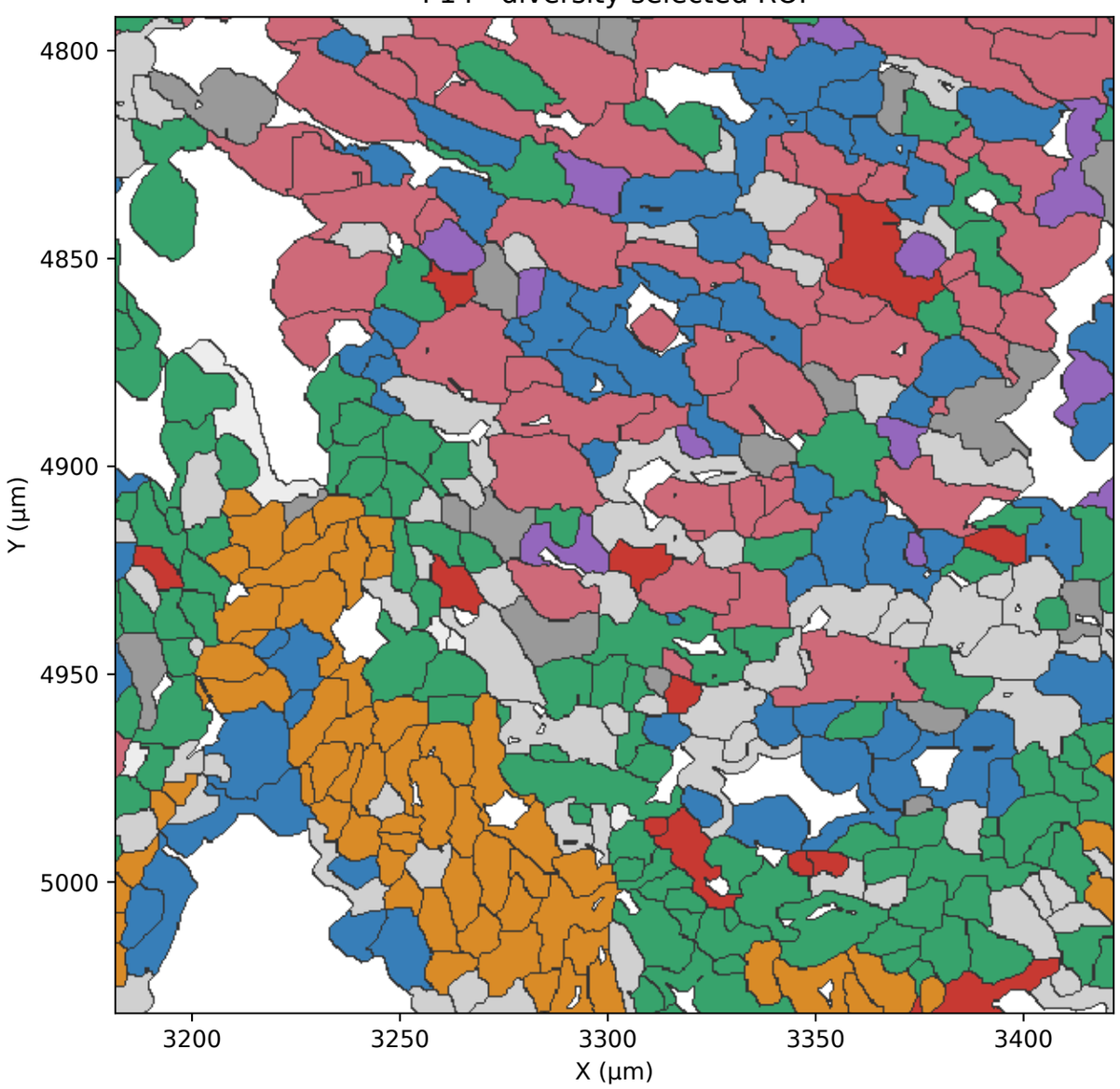


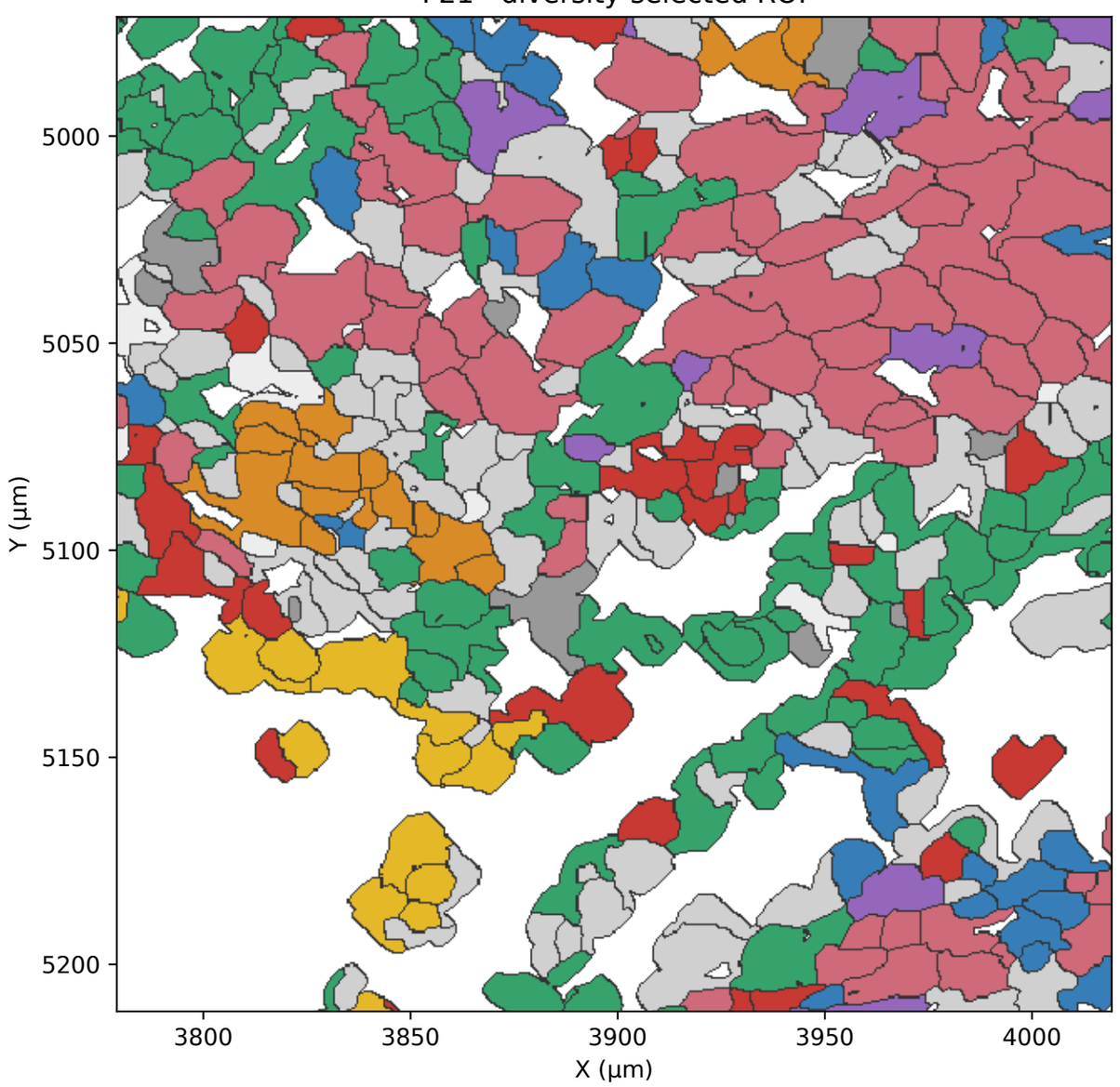

P21 · supported finer identities on original coordinates
Gray includes unresolved fine identity and other cardiac profiles; sparse categories are not absent-cell claims
Prioritized CM candidates · finer labels not forced
P21_M3: Regional electrical candidate
Strict FB cell+nuclear family assignments
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Strict EC cell+nuclear subtype assignments
Capillary-like
Endocardial-like
Arterial-like
Venous-like
Lymphatic-like
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000
6000

Segmentation-aware spatial context · boundaries and labels
Antibody-guided refinement retains deposited IDs; neither these boundaries nor labels are independent anatomical truth

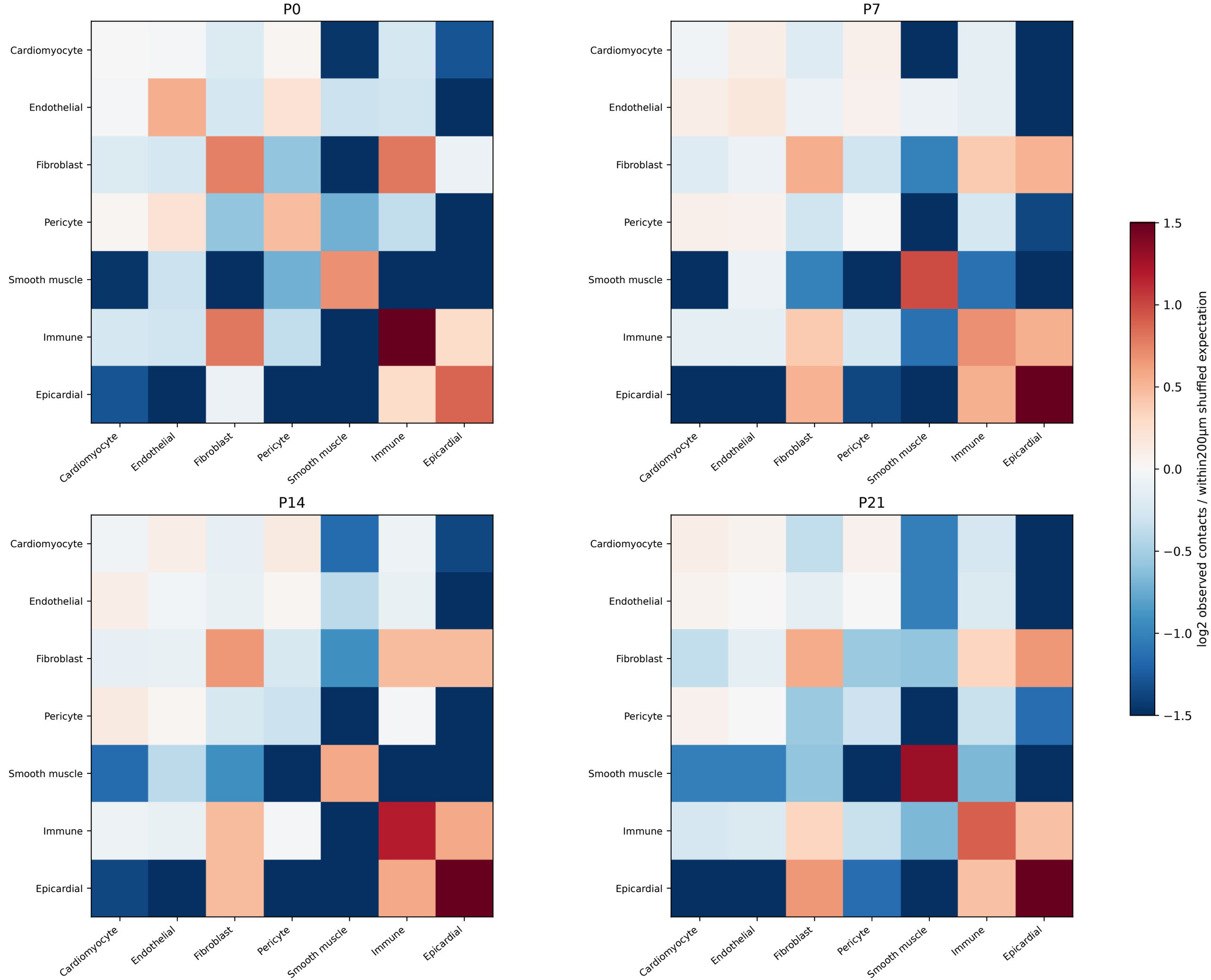


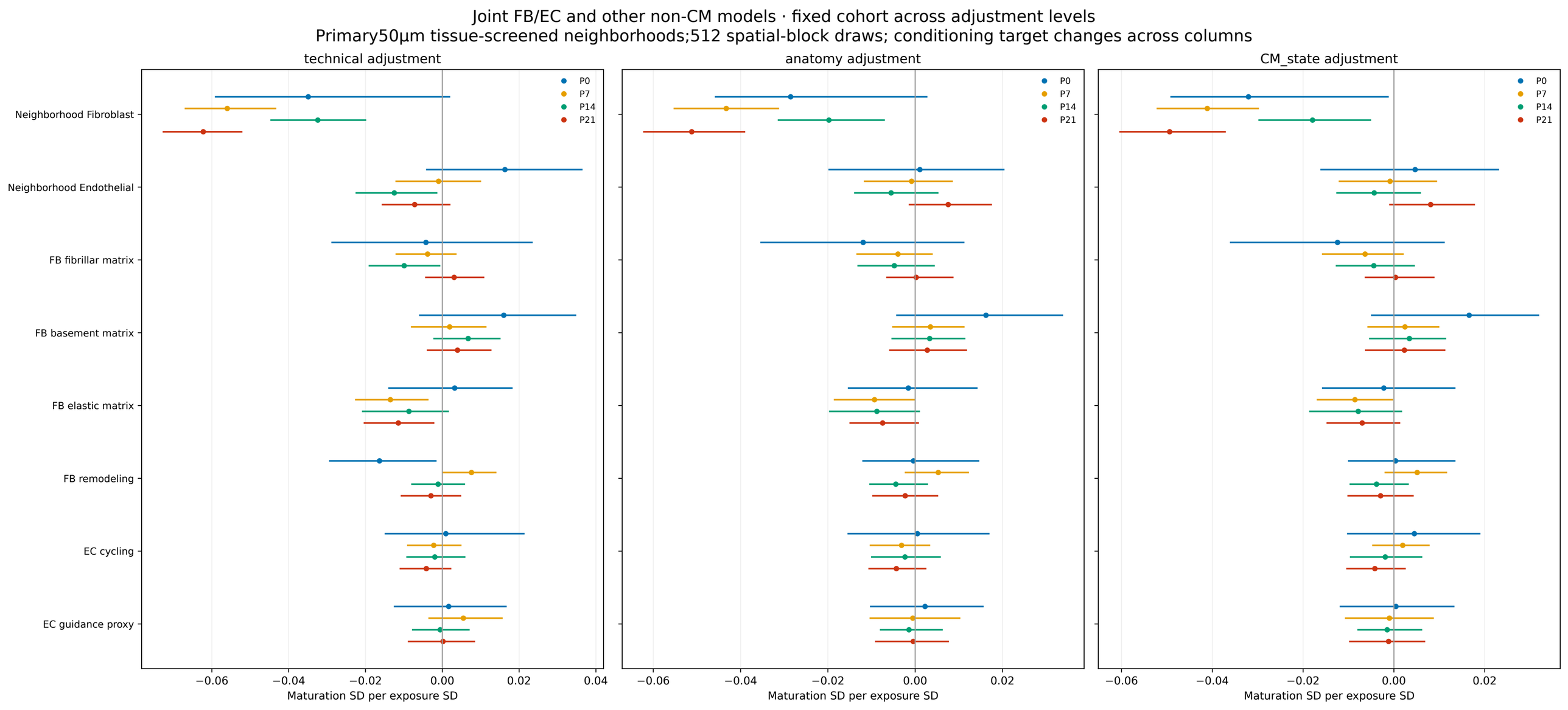


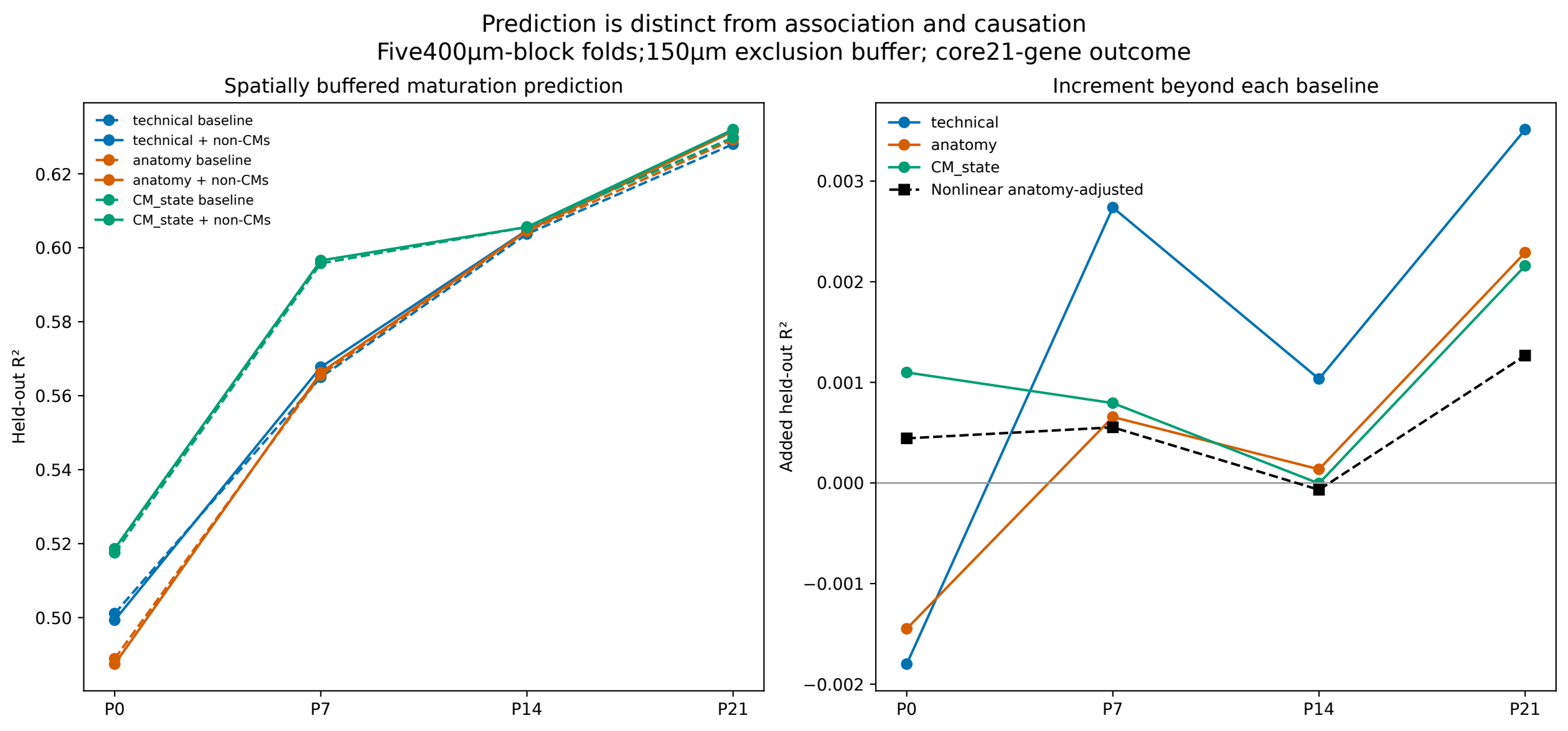




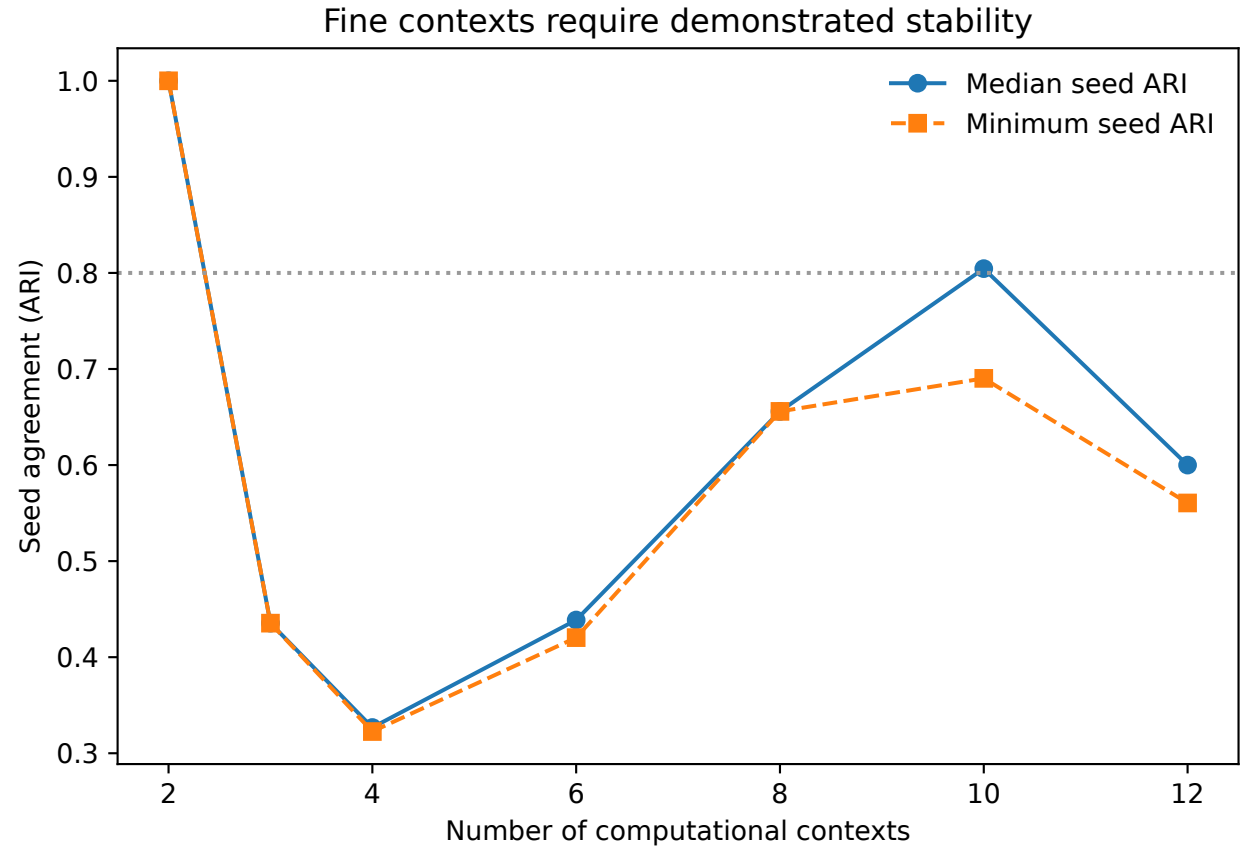
Cell-distribution hypotheses after preserving local composition
199 within-section label permutations; conditional co-occurrence, not communication or animal replication

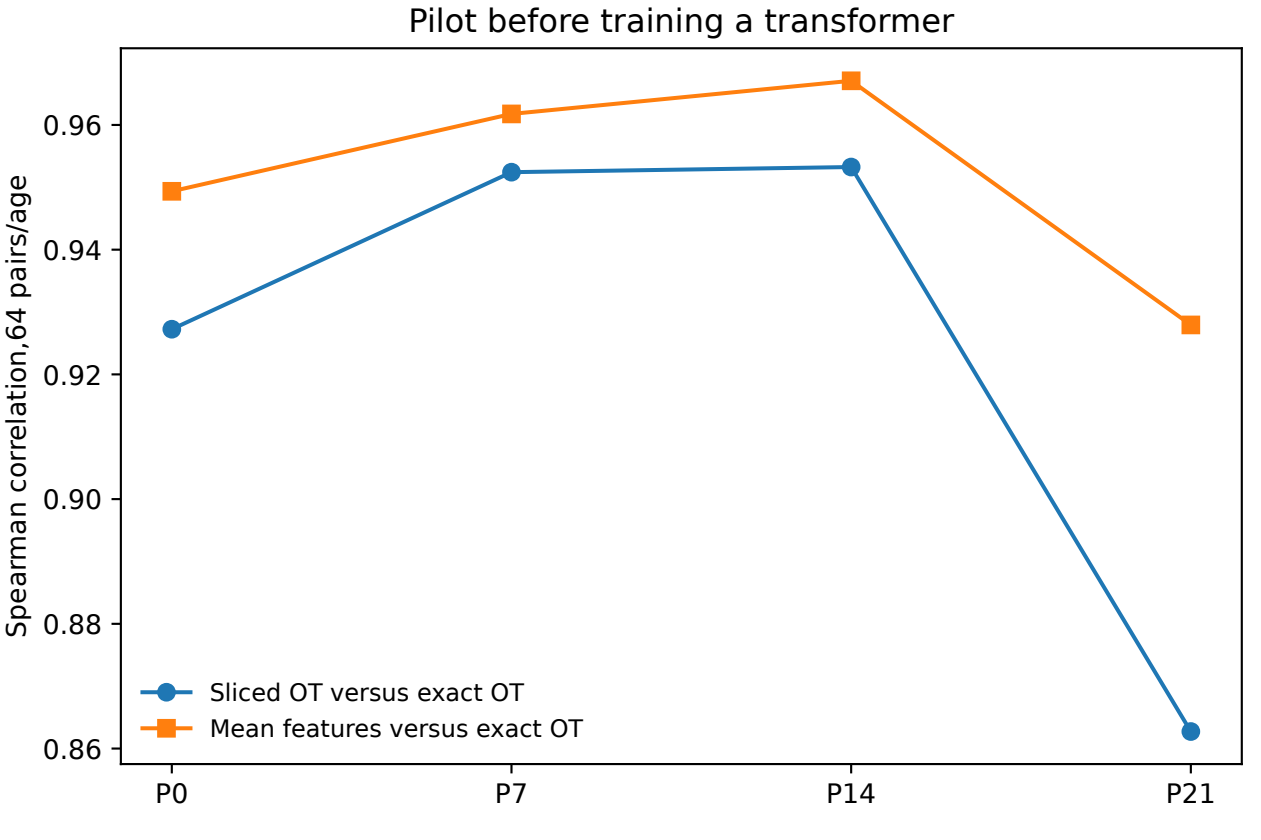



Joint FB/EC and other non-CM models · fixed cohort across adjustment levels
Primary50µm tissue-screened neighborhoods;512 spatial-block draws; conditioning target changes across columns
technical adjustment
anatomy adjustment
CM_state adjustment
P0
P7
P14
P21
Neighborhood Fibroblast
Neighborhood Endothelial
FB fibrillar matrix
FB basement matrix
FB elastic matrix
FB remodeling
EC cycling
EC guidance proxy
−0.06
−0.04
−0.02
0.00
0.02
0.04
Maturation SD per exposure SD


Prediction is distinct from association and causation
Five400µm-block folds;150µm exclusion buffer; core21-gene outcome
Spatially buffered maturation prediction
Held-out R²
technical baseline
technical + non-CMs
anatomy baseline
anatomy + non-CMs
CM_state baseline
CM_state + non-CMs
0.62
0.60
0.58
0.56
0.54
0.52
0.50
P0
P7
P14
P21
Increment beyond each baseline
Added held-out R²
technical
anatomy
CM_state
Nonlinear anatomy-adjusted
0.003
0.002
0.001
0.000
−0.001
−0.002
P0
P7
P14
P21

Method choice is evidence-based
CellCharter/Wormhole optional; pilot uses observed non-CM type/program distributions and tests distances only





Findings and interpretation gates

All408,054 original profiles remain accessible. Seven broad resolved lineages and conservative finer CM/FB/EC candidates are carried forward with explicit uncertainty.

Niche selection:K=2. Seed-stable computational niches; not independently validated anatomy. These are graph-context classes, not new cell identities or manually validated cardiac compartments.

Contact-enrichment patterns include concentrated smooth-muscle, epicardial and immune neighborhoods. Local-shuffle comparisons distinguish these from a simple global abundance contrast.

Joint primary models use measured non-CM RNA on one CM cohort at each age. EC cycling/guidance do not show a consistent negative relationship with the overall maturation panel acrossP0/P7.

Higher local fibroblast fraction has a small negative maturation association at later stages. It is a relative-composition contrast, not proof that fibroblasts suppress maturation; individual FB matrix programs add weaker evidence.

1 stage/exposure relationships meet the predeclared operational robustness screen (effect≥0.05SD, interval direction, native/balanced/Myl2-withheld score agreement, paired geometry agreement, technical-model agreement, low collinearity and conservative-QC support).

The screen prioritizes follow-up within these sections. It is not a multiplicity-corrected discovery procedure or independently replicated validation. All tested effects and failed gates are exported.

Candidate signaling tables use a small curated ligand/receptor-component list and opposite-library RNA expression support. Receptor complexes, protein abundance, activation and causal effects are not established.

CellCharter could provide a niche-backend sensitivity. Wormhole could scale comparisons of richer neighborhood distributions. Neither fixes missing genes, uncertain segmentation, age/batch confounding or lack of biological replication.

Next experimental priorities: independent age-matched sections/animals, histological chamber/compact-trabecular annotations, selected boundary review, and targeted perturbation of a nominated non-CM pathway with CM maturation readouts.